Wasimul Karim

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Research Interests

My research focuses on areas such as **Machine Learning**, **Computer Vision**, and **Multi-Modal Collaborative AI Systems**. I aim to integrate computational methods with real-world applications to advance both scientific understanding and practical impact.

Education

United International University February 2021 - July 2025
Bachelor of Computer Science and Engineering
CGPA: 3.87/4.00
Awarded *Magna Cum Laude* for Academic Excellence
CGPA in Major: 3.89/4.00

Work Experience

- **Research Assistant** October 2026 - Present
  Department of Computer Science and Engineering,
  United International University, Dhaka, Bangladesh

- **Research Assistant** October 2025 - Present
  Department of Computer Science and Engineering,
  University of Scholars, Dhaka, Bangladesh

  Responsibilities:

  - Conduct research on deep learning and multimodal AI, including model development, experimentation, and performance evaluation.
  - Assist in preparing manuscripts, datasets, and experimental pipelines, while collaborating with faculty and peers to support ongoing research projects.

- **Undergraduate Teaching Assistant** June 2023 - July 2025
  Department of Computer Science and Engineering,
  United International University, Dhaka, Bangladesh
  Taken over:

  - *Introduction to Computer Systems* — 3 Trimesters
  - *Structured Programming Language* — 3 Trimesters
  - *Object Oriented Programming* — 3 Trimesters
  - *Advanced Object Oriented Programming* — 1 Trimester
  - *Data Structure and Algorithms I* — 3 Trimesters
  - *Data Structure and Algorithms II* — 2 Trimesters

  Responsibilities:

  - Assisted in preparing and delivering lectures, tutorials, and lab sessions.

- Provided academic support and guidance to students, including counseling and conducting Q&A sessions.

Research Experience

- **Graduate Researcher** December 2023 - Present
  Applied Artificial Intelligence and INtelligent Systems Laboratory,
  Under the supervision of Prof. Dr. Sami Azam, Charles Darwin University (CDU), Australia.
  - Develop advanced multimodal and deep learning methods for medical imaging, sports analytics, digital media forensics, and intelligent visual understanding.
  - Design AI-driven solutions for sustainable and industrial applications, including hydrogen technologies, smart grids, waste management, autonomous UAV systems, and optimization.
  - Investigate trustworthy and scientific AI through generative AI, multimodal learning, and bioinformatics, with an emphasis on robust, interpretable, and real-world AI systems.

Peer Reviewed

[1] **Wasimul Karim**, Md. Abdur Rahman, Mohaimenul Azam Khan Raiaan, Tamanna Shermin, Md Rafiqul Islam, Mukhtar Hussain, Sami Azam *"A Fine-Grained Attention and Geometric Correspondence Model for Musculoskeletal Risk Classification in Athletes Using Multimodal Visual and Skeletal Features"* 2026, *Computers and Electrical Engineering*, Published - DOI: 10.1016/j.compeleceng.2026.111281

[2] **Wasimul Karim**, Riasad Alvi, Sayeem Been Zaman, Arefin Ittesafun Abian, Mohaimenul Azam Khan Raiaan, Saddam Mukta, Rafi Ur Rashid, Md Rafiqul Islam, Yakub Sebastian, Sami Azam, *"Generative Artificial Intelligence in Bioinformatics: A Systematic Review of Models, Applications, and Methodological Advances"* 2026, *Archives of Computational Methods in Engineering*, Published - DOI: 10.1007/s11831-026-10743-z

[3] **Wasimul Karim**, Sayeem Been Zaman, Debopom Sutradhar, Ripon Kumar Debnath, Sami Azam, Kheng Cher Yeo, Mirjam Jonkman, *"MedSegNet-AXU: Advanced Segmentation and Graph-Based Classification of Multiple Sclerosis Lesions from 3D MRI Imaging via Radiomics and Clinical Data Fusion"* 2026, *Complex and Intelligent Systems*, Published - DOI: 10.1007/s40747-026-02535-6

[4] Md Adnanul Islam, **Wasimul Karim**, Md Mahbub Alam, Subhey Sadi Rahman, Md Abdur Rahman, Arefin Ittesafun Abian, Mohaimenul Azam Khan Raiaan, Kheng Cher Yeo, Deepika Mathur, Sami Azam, *"Learning to Weigh Waste: A Physics-Informed Multimodal Fusion Framework and Large-Scale Dataset for Commercial and Industrial Applications"* 2026, *IEEE Transactions on Big Data*, Status: Under Review - Manuscript ID: TBD-2026-03-0149, DOI: 10.48550/arXiv.2603.00931

[5] **Wasimul Karim**, Nur Mohammad Fahad, Abdul Hasib Siddique, Md Rafiqul Islam, Hooman Mehdijadeh-Rad, Asif Karim, Sami Azam, *"Multi-Modal Semantic Segmentation of Electrolyzer Components for Sustainable Hydrogen Technologies: A Dual-Branch Deep Learning Approach"* 2026, *The International Journal of Advanced Manufacturing Technology*, Status: Under Review - Manuscript ID: EMID:27129b4f1d75568b, DOI: 10.48550/arXiv.2607.16056

[6] **Wasimul Karim**, Sayeem Been Zaman, Arefin Ittesafun Abian, Reem E. Mohamed, Md Rafiqul Islam, Asif Karim, Sami Azam *"DeepAgent: A Dual Stream Multi Agent Fusion for Robust Multimodal Deepfake Detection"* 2026, *Multimedia Tools and Applications*, Status: Under Review - Manuscript ID: EMID:3e3abf657c228da5, DOI: 10.48550/arXiv.2512.07351

[7] **Wasimul Karim**, Md. Delwar Shahadat Deepu, Arefin Ittesafun Abian, Mohaimenul Azam Khan Raiaan, Sami Azam *"EG-UNet: Edge-Guided Dual-Path Segmentation for Pixel-Level Deepfake Localization"* 2026, *Multimedia Tools and Applications*, Status: Under Review - Manuscript ID: EMID:024dd83ba75cbe7e

[8] Nur Mohammad Fahad, **Wasimul Karim**, Md Al Amin Sarker, Abdul Hasib Siddique, Sami Azam, Suresh Thennadil, *"Optimizing performance and privacy in anomaly classification of smart grids: A deep gated residual*

network with robust feature selection and the integration of federated learning" 2026, *Cluster Computing*, Status: Under Review - Manuscript ID: 0b17336e-749a-4bed-82ac-ddb3dde9bb42

[9] Nur Mohammad Fahad, **Wasimul Karim**, Mohaimenul Azam Khan Raiaan, Ripon Kumar Debnath, Reem E. Mohamed, Sheikh Izzal Azid, Abdul Hasib Siddique, Sami Azam, *"Vision–Planning Framework for Autonomous UAV Inspection of Photovoltaic Plants with Mobile-SAM and Multi-Objective Discrete Particle Swarm Optimisation"* 2026, *Engineering Applications of Artificial Intelligence*, Status: Under Review, Manuscript ID: EAAI-26-21600

[10] Debopom Sutradhar, Nur Mohammad Fahad, **Wasimul Karim**, Mohaimenul Azam Khan Raiaan, Sami Azam, Saddam Mukta, Sheikh Izzal Azid, Hamish A. Campbell *"MO-SA: An Optimized Multi Objective Simulated Annealing Approach for Efficient Drone-Based Products Delivery"* 2026, *Discover Computing*, Status: Under Review - Manuscript ID: 1a3d3546-8482-452c-8034-d7610789289d

[11] Sayeem Been Zaman, **Wasimul Karim**, Raiyan Rahman, *"EffiBurn: Leveraging EfficientNet and Ensemble Learning for Burn Severity Classification,"* 2025, *International Conference on Electrical, Computer and Communication Engineering (ECCE-2025)*, Published - DOI: 10.1109/ECCE64574.2025.11014049

Achievements

Received merit-based tuition waivers for outstanding academic performance:

- 100% waiver — 2 Trimesters
- 50% waiver — 3 Trimesters
- 25% waiver — 4 Trimesters

Technical Skills

- **Packages:** Scikit-learn, Numpy, Pandas, OpenCV, Tensorflow, PyTorch
- **Programming:** Python, Java, C/C++, PHP, Javascript, Bash, LATEX
- **Database & Tools:** MySQL, MongoDB, Jira, Overleaf

Extra Curricular Activity

- **Advisor** May 2026 - Present
  Students' International Affairs Society, SIAS-UIU
  - Provide strategic guidance and mentorship to the student organization in planning activities, managing challenges, and achieving its goals.
  - Support student members in leadership development, teamwork, and professional growth through regular communication and constructive feedback.
- **General Secretary** November 2023 - September 2024
  Students' International Affairs Society, SIAS-UIU
  - Coordinated student activities and managed society events and communications.
  - Represented students in international affairs initiatives and collaborated with other organizations.
- **Organizer**
  i. Foreign Education Fair, UIU December 2021
  ii. Webinar on "A Road to Erasmus Mundus Master's Degree and Erasmus+ Scholarship" June 2023
  - Planned, organized, and executed events, ensuring smooth logistics and participant engagement.

- Promoted events through marketing and coordinated with speakers and attendees.

Referees

Dr. Sami Azam
Professor & Discipline Chair of IT
Faculty of Science and Technology,
Charles Darwin University, Australia
Email: sami.azam@cdu.edu.au

Mohaimenul Azam Khan Raiaan
Assistant Lecturer & PhD Student
Dept. Data Science & AI, Faculty of IT,
Monash University, Australia
Email: mohaimenul.raiaan@monash.edu